

Notes: Cornett, McNutt, Strahan, and Tehranian (JFE, 2011)

Liquidity Risk Management and Credit Supply in the Financial Crisis

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1 Big picture

- **Question.** How did banks manage liquidity risk during the 2007–2009 financial crisis, and how did their liquidity management affect the supply of credit?
- **Core empirical claim.** Banks with more stable funding sources, especially core deposits and equity capital, were better able to maintain lending during the crisis. Banks with more liquidity risk on the asset side or off-balance sheet reduced credit supply and accumulated liquid assets.
- **Key shock.** The paper uses the financial crisis as a liquidity shock, measured mainly by the TED spread and alternatively by a crisis-regime indicator for 2007Q3–2009Q2.
- **Liquidity-risk channels.** The paper separates liquidity risk into three components: illiquid assets on the balance sheet, unstable or stable funding on the liability side, and off-balance sheet loan commitments that can be drawn during stress.
- **Main result on liquid assets.** Banks exposed to higher liquidity risk respond to the crisis by increasing liquid assets. This response is especially strong for banks holding more illiquid assets and unused commitments.
- **Main result on credit supply.** Banks with higher core deposits and equity capital expand lending and credit supply relative to banks that rely more on wholesale funding. Banks with larger unused commitments and illiquid assets reduce new credit supply.
- **Large versus small banks.** Effects are generally stronger among large banks. This is economically important because large banks hold most of the banking system’s assets and commitments.
- **Macro implication.** The paper uses coefficient estimates to calculate how much more credit large banks would have supplied in 2008Q4 if liquidity exposure had been lower. The estimated magnitudes are large relative to observed credit contraction.
- **Economic interpretation.** The crisis did not simply reduce credit because all banks faced the same aggregate liquidity shock. Credit supply declined more at banks whose pre-crisis balance sheets made them vulnerable to liquidity pressure.
- **Takeaway.** Liquidity risk management is a bank-level mechanism linking financial-market stress to the real economy. Stable funding and balance-sheet liquidity determine whether banks absorb or amplify a market-wide liquidity shock.

2 Incremental contribution of the paper

2.1 Contribution relative to bank liquidity and financial crisis research

- Prior work documents that financial crises reduce credit and that aggregate liquidity dries up. This paper contributes by showing *which banks* contract credit more and why.
- Instead of relying only on aggregate time-series movements in liquidity, the paper exploits cross-sectional variation across banks in pre-crisis balance-sheet exposure.
- This cross-sectional design helps identify whether banks’ own liquidity positions shape their response to a common financial crisis shock.

Interpretation: The paper moves the discussion from “the banking system became illiquid” to “banks with particular funding and asset structures became constrained and reduced credit supply.”

2.2 Contribution relative to the traditional deposit-run view

- Traditional bank-run models focus on the withdrawal of demand deposits.

- The paper shows that modern bank liquidity risk is more complex. Core deposits are relatively stable during crises, while wholesale deposits, off-balance sheet commitments, and illiquid assets create liquidity pressure.
- This reframes liquidity risk management as a balance-sheet-wide problem rather than a retail-deposit withdrawal problem.

Interpretation: The paper updates classic banking theory for modern banks that rely on wholesale funding and loan commitments.

2.3 Contribution relative to credit-supply identification

- A key challenge in crisis research is separating loan demand from loan supply.
- The paper uses bank fixed effects, quarter fixed effects, and interactions of liquidity-risk variables with the TED spread or crisis indicator to isolate how banks' balance sheets affect changes in loans and credit supply.
- It also controls for off-balance sheet commitments and loan demand proxies in robustness tests.

Interpretation: The paper's empirical strategy is designed to show that reduced credit supply is not only borrower demand collapsing; it is also banks actively managing liquidity risk.

2.4 Contribution relative to policy debates

- The Federal Reserve expanded aggregate liquidity during the crisis, but credit supply still contracted.
- The paper helps explain why monetary policy and liquidity facilities may not translate into new credit if individual banks hoard liquidity because of their own exposures.
- The results support policy attention to stable funding, liquidity buffers, and off-balance sheet commitments.

Interpretation: The paper anticipates later regulatory emphasis on liquidity coverage, stable funding, and commitment exposure.

3 Data and measurement

3.1 Bank panel

- The paper uses quarterly Call Report data from 2006Q1 through 2009Q2.
- Banks are aggregated to the high-holder level so that commonly controlled institutions are treated as a single bank organization.
- Large banks are those with beginning-of-quarter assets above \$1 billion.
- Small banks are those with beginning-of-quarter assets below \$1 billion.
- The sample is unbalanced across quarters but contains thousands of bank-quarter observations.

Interpretation: The high-holder aggregation is important because liquidity risk and lending decisions often occur at the bank holding company level.

3.2 Main dependent variables

The paper studies three balance-sheet changes, each scaled by beginning-of-period assets or commitments:

$$\Delta LiquidAssets_{i,t}/Assets_{i,t-1}, \quad (1)$$

$$\Delta Loans_{i,t}/Assets_{i,t-1}, \quad (2)$$

$$\Delta CreditSupply_{i,t}/(Commit_{i,t-1} + Assets_{i,t-1}). \quad (3)$$

- **Liquid assets** include cash plus liquid securities.
- **Loans** capture on-balance sheet lending.
- **Credit supply** equals loans plus unused commitments, capturing both loans outstanding and precommitted credit capacity.

Interpretation: Credit supply is broader than loans. A bank can appear to expand loans because clients draw down commitments, while total new credit capacity may actually decline.

3.3 Liquidity-risk variables

- **Illiquid assets/assets:** loans and leases net of unearned income plus all MBS and ABS held for sale or investment. This captures asset-side illiquidity.
- **Core deposits/assets:** insured deposits and transaction deposits. This captures stable funding.
- **Capital/assets:** Tier 1 capital plus qualifying perpetual preferred stock. This captures loss-absorbing funding.
- **Commitments/(commitments+assets):** unused commitments to commitments plus assets. This captures off-balance sheet liquidity exposure.
- **Log assets:** controls for bank size.

Interpretation: The paper’s central logic is that these variables determine how exposed a bank is to the liquidity shock.

3.4 Crisis intensity variables

The main crisis intensity measure is the TED spread:

$$TED_t = LIBOR_t - TreasuryBill_t. \quad (4)$$

The alternative crisis indicator is:

$$CRISIS_t = 1\{2007Q3 \leq t \leq 2009Q2\}. \quad (5)$$

Interpretation: The TED spread measures stress in interbank and money markets. The crisis dummy captures the broader crisis regime.

4 Empirical specifications

4.1 TED-spread interaction model

The main fixed-effects specification is:

$$Y_{i,t} = \alpha_i + \lambda_t + \beta_1 \text{LiquidityRisk}_{i,t-1} + \beta_2 \text{LiquidityRisk}_{i,t-1} \times \text{TED}_t \quad (6)$$

$$+ \beta_3 \text{CoreDeposits}_{i,t-1} + \beta_4 \text{CoreDeposits}_{i,t-1} \times \text{TED}_t \quad (7)$$

$$+ \beta_5 \text{Capital}_{i,t-1} + \beta_6 \text{Capital}_{i,t-1} \times \text{TED}_t \quad (8)$$

$$+ \beta_7 \text{CommitExposure}_{i,t-1} + \beta_8 \text{CommitExposure}_{i,t-1} \times \text{TED}_t \quad (9)$$

$$+ \beta_9 \text{LogAssets}_{i,t-1} + \beta_{10} \text{LogAssets}_{i,t-1} \times \text{TED}_t + \varepsilon_{i,t}. \quad (10)$$

- $Y_{i,t}$ is one of the three outcomes: liquid assets, loans, or credit supply.
- α_i are bank fixed effects.
- λ_t are quarter fixed effects.
- The interaction terms ask whether balance-sheet characteristics matter more when liquidity stress rises.

Interpretation: The interaction terms are the key coefficients. They show whether preexisting exposure changes bank behavior when the TED spread spikes.

4.2 Crisis-indicator model

The paper also replaces the TED spread with a crisis dummy:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_k \beta_k X_{i,t-1}^k + \sum_k \gamma_k X_{i,t-1}^k \times \text{CRISIS}_t + \varepsilon_{i,t}. \quad (11)$$

Interpretation: This specification asks whether the same balance-sheet characteristics matter during the financial crisis as a whole, not only when the TED spread fluctuates.

4.3 Economic magnitude calculation

The paper uses estimated coefficients to simulate how credit supply would have changed in 2008Q4 if banks' liquidity exposure had been low.

For a given exposure variable X , the adjustment is approximately:

$$\text{Adjustment} = (X^{25th} - X^{Actual}) \times \text{Shock} \times \hat{\beta}_{X \times \text{Shock}} \times \text{ExposureBase}. \quad (12)$$

Interpretation: This turns regression coefficients into dollar values, showing the macroeconomic importance of bank liquidity exposure.

5 Identification and empirical credibility

5.1 Identification logic

- The crisis shock is common to all banks.
- Banks differ in pre-crisis exposure to illiquid assets, unstable funding, capital, and unused commitments.
- Bank fixed effects absorb time-invariant bank characteristics.
- Quarter fixed effects absorb aggregate credit demand and macro shocks.
- The interaction of bank characteristics with crisis stress identifies differential responses across banks.

Interpretation: The design asks: holding constant the aggregate crisis, which banks changed lending more because of their own liquidity-risk profiles?

5.2 Threats and responses

- **Loan demand.** Borrowers of some banks may have reduced credit demand more than borrowers of others. The paper addresses this with robustness tests that add loan-share-by-TED and state-by-TED controls.
- **Bank size.** Large banks and small banks may behave differently. The paper estimates separate samples and size bins.
- **Commitment drawdowns.** Increases in loans can reflect borrowers drawing existing commitments. The paper therefore studies total credit supply, not just loans.
- **Core deposit inflows.** Deposits may flow into banks during crises. The paper distinguishes stable core deposits from wholesale deposits and shows stable funding mitigates credit contraction.

Interpretation: The robustness tests reinforce that the results reflect bank credit supply decisions, not only borrower demand or mechanical drawdowns.

6 Tables and figures

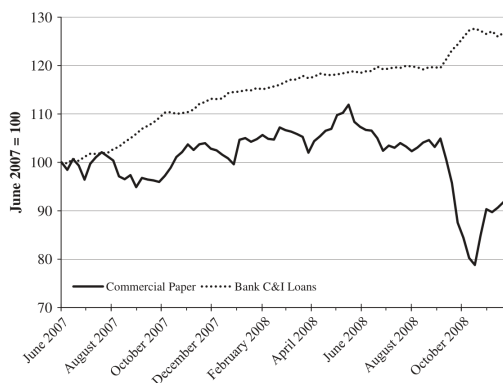
The figures and tables are directly cropped from the paper. Adjacent visuals are placed on the same row, and a single remaining visual is shown at width 0.6.

6.1 Figures 1 and 2: crisis liquidity background

Read:

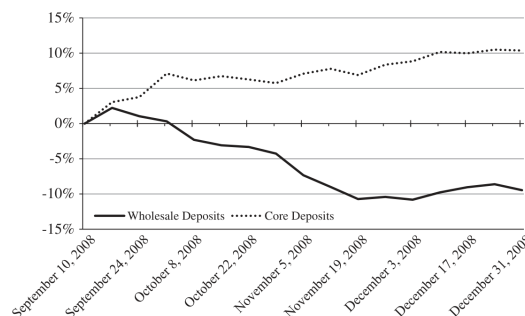
- Figure 1 shows that commercial paper outstanding collapses after Lehman, while bank C&I loans rise as firms draw down backup credit lines.
- Figure 2 shows wholesale deposits falling sharply after September 2008 while core deposits rise.
- Together, the figures show that modern banks face liquidity risk from wholesale funding and loan commitments rather than from retail core deposits.

Economic message: The figures motivate the paper's two main mechanisms: off-balance sheet liquidity risk materializes on balance sheet, and core deposits are a stabilizing funding source.



risers as commercial paper moves back on the balance sheet. This figure shows the growth of commercial paper and bank C&I loans through November 2008. Data are obtained from the website of the Board of Governors of the Federal Reserve (www.federalreserve.gov).

Figure 1: commercial paper and C&I loans



This figure shows the weekly percentage change in core and wholesale deposits at commercial banks from September 10, 2008, through November 2008. Core deposits include transactions deposits plus fully insured (< \$100,000) time deposits. Wholesale deposits include all other deposits. Data are obtained from the Federal Reserve's H8 weekly data on bank assets and liabilities.

not find such flows during the pre-crisis period. This liquidity risk increased their holdings of

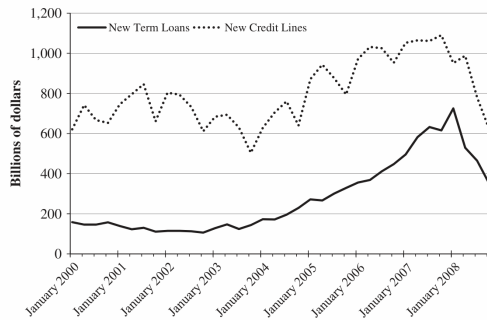
Figure 2: core and wholesale deposits

6.2 Figures 3 and 4: loan origination collapse and TED spread shock

Read:

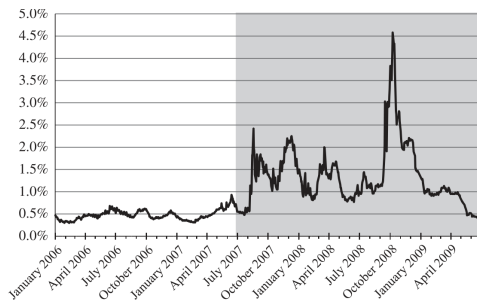
- Figure 3 shows that new term loans and credit lines issued to large businesses decline sharply during the crisis.
- Figure 4 shows the TED spread spiking during crisis periods, especially around late 2008.
- The shaded period in Figure 4 corresponds to the crisis regime used in the paper's dummy-variable specification.

Economic message: The figures connect bank credit supply with financial-market stress: loan originations collapse when liquidity stress peaks.



originations collapse. This figure shows the dollar value of new term loans and credit lines issued to large business (at the height of the financial crisis) through 2008. Data used to construct the figure are obtained from the database.

Figure 3: business loan originations



ad. This figure shows movements in the TED spread from 2006 through the second quarter of 2009. The TED spread is the three-month London Interbank Offered Rate (LIBOR) rate (obtained from the website of the Bulgarian National Bank) minus the three-month Treasury rate (from the Federal Reserve Economic Data (FRED) website of the Federal Reserve Bank of St. Louis). The shaded area includes the period we designate the crisis period in our empirical analysis.

Figure 4: TED spread

6.3 Tables 1 and 2: sample and summary statistics

Read:

- Table 1 reports the number of large and small commercial banks by quarter from 2006Q1 to 2009Q2.
- Table 2 shows changes in liquid assets, cash-like assets, loans, and credit supply for large and small banks in noncrisis quarters, crisis quarters, and 2008Q4.
- Large banks have higher commitments relative to commitments plus assets and lower reliance on core deposits than small banks.
- The crisis-period summary statistics show that banks adjust liquid assets and credit supply differently across size groups.

Economic message: The summary statistics show substantial cross-sectional variation in liquidity exposure, funding stability, and credit supply. This variation is essential for the paper's identification.

Table 1

Number of commercial banks examined between 2006Q1 and 2009Q2.

This table lists the distribution of the sample banks by quarter. We segregate banks into two size groups based on beginning of quarter book value of assets. Large banks are those banks with beginning of quarter assets greater than \$1 billion, and small banks are those banks with beginning of quarter assets less than \$1 billion. Bank asset size is collected from Federal Financial Institutions Examination Council *Call Reports of Condition and Income* found on the website of the Federal Reserve Bank of Chicago (www.chicagofed.org).

Year	Quarter	Large	Small	Total
2006	1	430	5,784	6,214
	2	432	5,789	6,221
	3	442	5,731	6,173
	4	434	5,563	5,997
2007	1	455	5,664	6,119
	2	453	5,670	6,123
	3	457	5,639	6,096
	4	462	5,539	6,001
2008	1	459	5,460	5,919
	2	483	5,583	6,066
	3	485	5,575	6,060
	4	468	5,440	5,908
2009	1	491	5,506	5,997

Table 1: sample counts

Table 2

Summary statistics for commercial bank characteristics and select cash flow statement items over the period 2006Q1 through 2009Q2. This table reports summary statistics for changes in liquid assets, loans, and credit supply at commercial banks and bank characteristics. Growth variables are standardized by beginning of period total assets or beginning of period assets plus unused commitments. Bank characteristics are beginning of period values. The data are observed quarterly for a panel of all US commercial banks over the period 2006Q1 through 2009Q2. Crisis quarters cover 2007Q3–2009Q2. Bank data, aggregated at the high holding company as appropriate, are from the *Call Reports of Condition and Income* accessed via the Federal Reserve Bank of Chicago website (www.chicagofed.org). Tests for significant differences in means are based on unpaired t-tests. Tests for significant differences between large bank and small bank medians are based on the Wilcoxon-Mann-Whitney test statistic. ***, **, and * denote that the large bank sample differs significantly from the small bank sample at the 1%, 5%, and 10% level, respectively.

	Average	25th percentile	Median	75th percentile	Standard deviation
Panel A: Large banks during noncrisis quarters					
Δ Liquid Assets/Assets (percent)	-0.18***	-1.17	-0.09***	0.99	2.52
Δ (Cash + US Treas + Fed Funds + Repos)/Assets (percent)	-0.10***	-0.86	-0.01***	0.76	2.51
Δ Loans/Assets (percent)	1.39	0.29	1.33***	2.57	3.03
Δ Credit Supply/(Commit + Assets) (percent)	1.60***	0.22	1.46***	2.91	3.63
Panel B: Small banks during noncrisis quarters					
Δ Liquid Assets/Assets (percent)	-0.40	-2.17	-0.21	1.64	4.11
Δ (Cash + US Treas + Fed Funds + Repos)/Assets (percent)	-0.36	-1.95	-0.15	1.49	4.12
Δ Loans/Assets (percent)	1.32	-0.31	1.06	2.72	3.33
Δ Credit Supply/(Commit + Assets) (percent)	1.41	-0.38	1.06	2.89	3.97
Panel C: Large banks during crisis quarters					
Δ Liquid Assets/Assets (percent)	-0.22*	-1.41	-0.20	1.05	3.39
Δ (Cash + US Treas + Fed Funds + Repos)/Assets (percent)	0.02	-0.88	-0.02	0.92	3.10
Δ Loans/Assets (percent)	0.85***	-0.40	0.83	2.13	3.39
Δ Credit Supply/(Commit + Assets) (percent)	0.50***	-0.90	0.54***	1.98	4.08
Panel D: Small banks during crisis quarters					
Δ Liquid Assets/Assets (percent)	-0.34	-2.30	-0.21	1.91	4.43
Δ (Cash + US Treas + Fed Funds + Repos)/Assets (percent)	-0.07	-1.84	0.01	1.96	4.41
Δ Loans/Assets (percent)	1.04	-0.60	0.76	2.44	3.39
Δ Credit Supply/(Commit + Assets) (percent)	0.96	-0.80	0.65	2.40	4.98
Panel E: Large banks during 2008Q4					
Δ Liquid Assets/Assets (percent)	0.54**	-0.81	0.15	1.79	3.09
Δ (Cash + US Treas + Fed Funds + Repos)/Assets (percent)	0.47*	-0.52	0.12	1.33	2.72
Δ Loans/Assets (percent)	0.54***	-0.32	0.75	1.94	4.40
Δ Credit Supply/(Commit + Assets) (percent)	0.07***	-1.16	0.20***	1.46	5.01
Panel F: Small banks during 2008Q4					
Δ Liquid Assets/Assets (percent)	0.16	-1.65	0.16	2.22	4.41
Δ (Cash + US Treas + Fed Funds + Repos)/Assets (percent)	0.14	-1.50	0.18	2.12	4.39
Δ Loans/Assets (percent)	0.97	-0.53	0.81	2.40	3.22
Δ Credit Supply/(Commit + Assets) (percent)	0.73	-0.83	0.56	2.20	3.57
Panel G: Large bank characteristics					
Illiquid Assets/Assets (percent)	77.88***	74.36	79.95***	84.82	11.41
Core Deposits/Assets (percent)	58.57***	52.97	61.08***	67.80	14.41
Capital/Assets (percent)	11.69***	9.56	10.39***	11.96	5.54
Commit/(Commit + Assets) (percent)	16.79***	10.89	15.22***	19.84	11.10
Book Value of Total Assets (\$MM)	19,739.53***	1,378.79	2,135.22***	5,361.55	116,421.91
Panel H: Small bank characteristics					
Illiquid Assets/Assets (percent)	70.27	63.09	73.34	80.86	15.10
Core Deposits/Assets (percent)	66.44	60.86	67.88	74.08	11.63
Capital/Assets (percent)	18.56	11.12	13.71	18.43	104.65
Commit/(Commit + Assets) (percent)	9.18	4.87	8.34	12.24	7.02
Book Value of Total Assets (\$MM)	195.39	59.08	123.22	254.98	197.35

Table 2: summary statistics

6.4 Tables 3 and 4: baseline fixed-effects results

Read:

- Table 3 interacts bank balance-sheet variables with the TED spread.
- For large banks, higher illiquid assets interacted with the TED spread predict more liquid-asset accumulation and lower credit supply.
- Higher core deposits and capital interacted with the TED spread predict stronger loan growth and credit supply.
- Higher commitments interacted with the TED spread predict lower credit supply, consistent with takedowns displacing new lending capacity.
- Table 4 repeats the analysis using the crisis indicator. The sign pattern is broadly consistent with Table 3.

Economic message: The tables are the paper's core evidence: banks exposed to more liquidity risk hoard liquidity and restrict credit, while banks with stable funding and capital continue lending.

Table 3

Fixed effects regressions of liquid asset, loan, and credit supply growth on TED spread, firm characteristics, and interactions.

This table reports fixed effects regressions of quarterly growth in liquid assets standardized by beginning of period assets. The table also reports fixed effects regressions of growth in loans standardized by beginning of period assets and growth in credit supply (i.e., growth in loans plus growth in unused commitments) standardized by beginning of period assets plus unused commitments. The data are observed quarterly for a panel of US commercial banks over the period 2000Q1 through 2009Q2. Large banks are those banks with beginning of quarter assets greater than \$1 billion, and small banks are those banks with beginning of quarter assets less than \$1 billion. Commercial bank data, aggregated at the high holding company as appropriate, are from *Call Reports of Condition and Income* accessed via the Federal Reserve Bank of Chicago website (www.chicagofed.org). Banks with asset growth greater than 10% during a quarter are dropped during that quarter to mitigate the effect of large mergers on changes in liquid assets, loans, and credit supply. Variables are winsorized at the 1st and 99th percentiles. TED spread is the quarterly average of the daily difference between the three-month London Interbank Offered Rate (LIBOR) and the three-month US Treasury bill secondary market rate. LIBOR data are from the Bulgarian National Bank website (www.bnb.bg/en) and Treasury bill data are from the Federal Reserve Economic Data (FRED) website of the Federal Reserve Bank of St. Louis (<http://research.stlouisfed.org/fred2/>). Standard errors, clustered at the firm level, are reported in parentheses. ***, **, and * denote that the coefficients are statistically significantly different from zero at the 1%, 5%, and 10% level, respectively.

	Δ Liquid Assets _{t-1} /Assets _{t-1}	Δ (Cash + US Treas + Fed Funds + Repos _{t-1})/Assets _{t-1}	Δ Loans _{t-1} /Assets _{t-1}	Δ (Credit _{t-1} /(Commit + Assets) _{t-1})
	(1)	(2)	(3)	(4)
Panel A: Large banks (total assets > \$1 billion)				
Illiquid Assets/Assets _{t-1}	0.209*** (0.023)	0.187*** (0.020)	-0.015 (0.014)	0.030* (0.017)
Illiquid Assets/Assets _{t-1} * TED	2.423*** (0.883)	0.773 (0.710)	-1.145* (0.678)	-1.340* (0.762)
Core Deposits/Assets _{t-1}	0.004 (0.014)	0.004 (0.010)	-0.025*** (0.009)	-0.044*** (0.011)
Core Deposits/Assets _{t-1} * TED	-0.837 (0.663)	-0.440 (0.601)	1.231*** (0.453)	1.390** (0.555)
Capital/Assets _{t-1}	-0.042 (0.050)	-0.096** (0.039)	0.042 (0.037)	0.088* (0.054)
Capital/Assets _{t-1} * TED	-0.499 (2.233)	-2.314 (2.068)	5.732** (2.637)	5.984** (2.699)
Commit/(Commit + Assets) _{t-1}	0.070*** (0.023)	-0.065*** (0.018)	0.176*** (0.027)	-0.012 (0.032)
Commit/(Commit + Assets) _{t-1} * TED	0.933 (0.961)	0.602 (0.736)	2.303** (0.937)	-2.753*** (0.855)
Log Assets _{t-1}	-0.016*** (0.005)	-0.017*** (0.004)	-0.017*** (0.005)	-0.027*** (0.007)
Log Assets _{t-1} * TED	0.186*** (0.072)	0.132** (0.061)	0.064 (0.052)	0.131* (0.067)
Firm dummies	Yes	Yes	Yes	Yes
Quarterly time dummies	Yes	Yes	Yes	Yes
N	6,462	6,462	6,462	6,462
Adjusted R-squared	0.244	0.205	0.404	0.342
Panel B: Small banks (total assets ≤ \$1 billion)				
Illiquid Assets/Assets _{t-1}	0.291*** (0.011)	0.278*** (0.007)	-0.067*** (0.004)	-0.051*** (0.006)
Illiquid Assets/Assets _{t-1} * TED	0.240 (0.296)	-0.702*** (0.265)	-0.546*** (0.163)	-0.516*** (0.183)
Core Deposits/Assets _{t-1}	-0.028*** (0.006)	-0.039*** (0.005)	0.002 (0.005)	-0.002 (0.006)
Core Deposits/Assets _{t-1} * TED	-0.229 (0.295)	-0.871*** (0.268)	0.673*** (0.204)	1.553*** (0.238)
Capital/Assets _{t-1}	-0.008 (0.012)	-0.006 (0.013)	0.070*** (0.009)	0.067*** (0.012)
Capital/Assets _{t-1} * TED	-1.456*** (0.590)	-2.744*** (0.471)	1.135*** (0.273)	1.058*** (0.325)
Commit/(Commit + Assets) _{t-1}	-0.167*** (0.010)	-0.152*** (0.010)	0.295*** (0.009)	-0.137*** (0.012)
Commit/(Commit + Assets) _{t-1} * TED	3.090*** (0.611)	1.933*** (0.568)	1.071** (0.459)	-3.034*** (0.557)
Log Assets _{t-1}	-0.040*** (0.003)	-0.038*** (0.003)	-0.015*** (0.004)	-0.028*** (0.006)
Log Assets _{t-1} * TED	-0.311*** (0.033)	-0.269*** (0.029)	0.079*** (0.022)	0.021 (0.024)
Firm dummies	Yes	Yes	Yes	Yes
Quarterly time dummies	Yes	Yes	Yes	Yes
N	78,581	78,581	78,581	78,581
Adjusted R-squared	0.270	0.253	0.380	0.299

Table 3: TED-spread interactions

Table 4

Fixed effects regressions of liquid asset, loan, and credit supply growth on crisis indicator, firm characteristics, and interactions.

This table reports fixed effects regressions of quarterly growth in liquid assets standardized by beginning of period assets. The table also reports fixed effects regressions of growth in loans standardized by beginning of period assets and growth in credit supply (i.e., growth in loans plus growth in unused commitments) standardized by beginning of period assets plus unused commitments. The data are observed quarterly for a panel of US commercial banks over the period 2000Q1 through 2009Q2. Large banks are those banks with beginning of quarter assets greater than \$1 billion, and small banks are those banks with beginning of quarter assets less than \$1 billion. Commercial bank data, aggregated at the high holding company as appropriate, are from *Call Reports of Condition and Income* accessed via the Federal Reserve Bank of Chicago website (www.chicagofed.org). Banks with asset growth greater than 10% during a quarter are dropped during that quarter to mitigate the effect of large mergers on changes in liquid assets, loans, and credit supply. Variables are winsorized at the 1st and 99th percentiles. CRISIS is an indicator variable that takes the value of one for observations that occur during the period 2007Q3 through 2009Q2 and zero otherwise. Standard errors, clustered at the firm level, are reported in parentheses. ***, **, and * denote that the coefficients are statistically significantly different from zero at the 1%, 5%, and 10% level, respectively.

	Δ Liquid Assets/Assets _{t-1}	Δ (Cash + US Treas + Fed Funds + Repos _{t-1})/Assets _{t-1}	Δ Loans/Assets _{t-1}	Δ (Credit _{t-1} /(Commit + Assets) _{t-1})
	(1)	(2)	(3)	(4)
Panel A: Large banks (total assets > \$1 billion)				
Illiquid Assets/Assets _{t-1}	0.205*** (0.022)	0.180*** (0.019)	-0.012 (0.013)	0.020* (0.017)
Illiquid Assets/Assets _{t-1} * CRISIS	0.048*** (0.012)	0.023** (0.009)	-0.021*** (0.008)	-0.020** (0.009)
Core Deposits/Assets _{t-1}	-0.002 (0.012)	-0.002 (0.008)	-0.022*** (0.008)	-0.041*** (0.011)
Core Deposits/Assets _{t-1} * CRISIS	-0.002 (0.008)	0.001 (0.007)	0.020*** (0.006)	0.021*** (0.007)
Capital/Assets _{t-1}	-0.065 (0.048)	-0.123*** (0.038)	0.064* (0.035)	0.109* (0.050)
Capital/Assets _{t-1} * CRISIS	0.033 (0.030)	-0.010 (0.026)	0.063* (0.022)	0.072** (0.033)
Commit/(Commit + Assets) _{t-1}	-0.069*** (0.022)	-0.064*** (0.018)	0.189*** (0.025)	-0.020 (0.031)
Commit/(Commit + Assets) _{t-1} * CRISIS	0.029* (0.011)	0.013 (0.010)	0.014 (0.012)	-0.039*** (0.011)
Log Assets _{t-1}	-0.016*** (0.005)	-0.017*** (0.004)	-0.017*** (0.005)	-0.025*** (0.007)
Log Assets _{t-1} * CRISIS	0.002** (0.001)	0.001 (0.001)	0.002*** (0.001)	0.002*** (0.001)
Firm dummies	Yes	Yes	Yes	Yes
Quarterly time dummies	Yes	Yes	Yes	Yes
N	6,462	6,462	6,462	6,462
Adjusted R-squared	0.246	0.205	0.407	0.346
Panel B: Small banks (total assets < \$1 billion)				
Illiquid Assets/Assets _{t-1}	0.286*** (0.007)	0.274*** (0.007)	-0.068*** (0.004)	-0.049*** (0.005)
Illiquid Assets/Assets _{t-1} * CRISIS	0.014*** (0.003)	-0.003 (0.003)	-0.009*** (0.002)	-0.012*** (0.002)
Core Deposits/Assets _{t-1}	-0.032*** (0.005)	-0.044*** (0.005)	0.005 (0.005)	0.001 (0.006)
Core Deposits/Assets _{t-1} * CRISIS	0.003 (0.004)	-0.008** (0.003)	0.007*** (0.003)	0.022*** (0.003)
Capital/Assets _{t-1}	-0.016 (0.011)	-0.015 (0.012)	0.073*** (0.009)	0.071*** (0.011)
Capital/Assets _{t-1} * CRISIS	-0.008 (0.006)	-0.029*** (0.006)	0.017*** (0.004)	0.021*** (0.004)
Commit/(Commit + Assets) _{t-1}	-0.159*** (0.010)	-0.150*** (0.009)	0.300*** (0.009)	-0.150*** (0.011)
Commit/(Commit + Assets) _{t-1} * CRISIS	0.038*** (0.007)	0.029*** (0.007)	0.004 (0.005)	-0.046*** (0.007)
Log Assets _{t-1}	-0.041*** (0.003)	-0.040*** (0.003)	-0.014*** (0.004)	-0.025*** (0.005)
Log Assets _{t-1} * CRISIS	-0.004*** (0.000)	-0.003*** (0.000)	0.001*** (0.000)	0.000 (0.000)
Firm dummies	Yes	Yes	Yes	Yes
Quarterly time dummies	Yes	Yes	Yes	Yes
N	78,581	78,581	78,581	78,581
Adjusted R-squared	0.270	0.253	0.381	0.302

Table 4: crisis-indicator interactions

6.5 Tables 5 and 6: size bins and economic magnitude

Read:

- Table 5 shows that liquidity-risk effects differ across bank size bins.
- The largest banks show the strongest interaction effects, which matters because they account for most banking system assets and commitments.
- Table 6 translates coefficients into dollar effects for 2008Q4.
- If large banks had low liquidity exposure, simulated credit contraction would have been dramatically smaller.

Economic message: The estimates are not only statistically significant; they imply large reductions in aggregate credit production during the crisis.

Table 5

Fixed effects regressions of credit growth on TED spread, firm characteristics, and interactions: standardized regression coefficients.

This table reports fixed effects regressions of quarterly growth in credit supply (i.e., growth in loans plus growth in unused commitments), standardized by beginning of period assets plus unused commitments. The data are observed quarterly for a panel of US commercial banks over the period 2006Q1 through 2009Q2. Commercial bank data, aggregated at the high holding company as appropriate, are from Call Reports of Condition and Income accessed via the Federal Reserve Bank of Chicago website (www.chicagofed.org). Banks with asset growth greater than 10% during a quarter are dropped during that quarter to mitigate the effect of large mergers on changes in liquid assets, loans, and credit supply. Variables are winsorized at the 1st and 99th percentiles. TED spread is the quarterly average of the daily difference between the three-month London Interbank Offered Rate (LIBOR) and the three-month US Treasury bill secondary market rate. LIBOR data are from the Bulgarian National Bank website (www.bnb.bg/#) and Treasury bill data are from the Federal Reserve Economic Data (FRED) website of the Federal Reserve Bank of St. Louis (<http://research.stlouisfed.org/fred2/>). Standard errors, clustered at the firm level, are reported in parentheses. ***, **, and * denote that the coefficients are statistically significantly different from zero at the 1%, 5%, and 10% level, respectively.

	Large banks (total assets > \$1 billion)	Medium banks (total assets \$500 million to \$1 billion)	Small banks (total assets \$100 million to \$500 million)	Smallest banks (total assets ≤ \$100 million)
	(1)	(2)	(3)	(4)
<i>Illiquid Assets/Assets_{t-1}</i>	0.116* (0.068)	-0.111** (0.055)	-0.180*** (0.028)	-0.383*** (0.033)
<i>Liquid Assets/Assets_{t-1} * TED</i>	-5.253* (2.980)	-5.581*** (2.146)	-2.235** (1.140)	-1.493 (1.187)
<i>Core Deposits/Assets_{t-1}</i>	-0.230*** (0.058)	-0.117** (0.048)	-0.043* (0.026)	0.037 (0.034)
<i>Core Deposits/Assets_{t-1} * TED</i>	7.050** (2.886)	6.608*** (2.523)	4.565*** (1.045)	3.929*** (1.259)
<i>Capital/Assets_{t-1}</i>	0.134* (0.082)	0.056 (0.104)	0.233*** (0.040)	0.017 (0.044)
<i>Capital/Assets_{t-1} * TED</i>	9.120** (4.113)	-0.612 (2.315)	2.349** (1.081)	3.024* (1.605)
<i>Commit/(Commit + Assets)_{t-1}</i>	-0.048 (0.123)	-0.148** (0.070)	-0.171*** (0.028)	-0.421*** (0.028)
<i>Commit/(Commit + Assets)_{t-1} * TED</i>	-10.702*** (3.322)	-4.714* (2.726)	-5.057*** (1.287)	-2.295* (1.254)
<i>Log Assets_{t-1}</i>	-1.246*** (0.318)	-0.325*** (0.060)	-0.641*** (0.048)	-0.708*** (0.073)
<i>Log Assets_{t-1} * TED</i>	6.091* (3.118)	-1.751 (2.091)	0.078 (0.985)	1.573 (1.121)
Firm dummies	Yes	Yes	Yes	Yes
Quarterly time dummies	Yes	Yes	Yes	Yes
N	6,462	7,097	38,166	33,318
Adjusted R-squared	0.342	0.347	0.300	0.367

Table 5: size-bin regressions

Table 6

Economic impact of liquidity shocks during 2008Q4.

This table reports the estimated effect of the liquidity shock in the financial crisis on changes in liquid assets, loans, and credit supply in 2008Q4. We adjust the actual changes in 2008Q4 as if each right-hand side variable observed below the 75th percentile (measured 2007Q4) had been equal to the 75th percentile level to simulate the response of the banking system if all banks had had low liquidity exposure. Panel A subjects banks to the 250 basis points TED spread shock observed in 2008Q4 and uses coefficients from Table 3; Panel B subjects banks to the overall shock of moving into the financial crisis regime and uses the coefficients from Table 4. ***, **, and * denote that the coefficient estimates are statistically significantly different from zero at the 1%, 5%, and 10% levels, respectively. The data are observed quarterly for a panel of US commercial banks over the period 2006Q1 through 2009Q2.

	$\Delta \text{Liquid Assets}_{2008Q4}$	$\Delta (\text{Cash} + \text{US Treas} + \text{Fed Funds} + \text{Repos})_{2008Q4}$	$\Delta \text{Loans}_{2008Q4}$	$\Delta \text{Credit}_{2008Q4}$
	(1)	(2)	(3)	(4)
<i>Panel A: Large banks (total assets > \$1 billion) (N=427), based on coefficients from Table 3</i>				
Actual change (\$Bn)	168	121	-52	-503
<i>Illiquid Assets</i> adjustment (\$Bn)	-10***	-3	5*	6*
<i>Core Deposits</i> adjustment (\$Bn)	-53	-28	79***	86**
<i>Capital</i> adjustment (\$Bn)	-2	-10	25**	26**
<i>Commit</i> adjustment (\$Bn)	-101	-65	-249**	297***
Adjusted change (\$Bn)	2	15	-192	-87
<i>Panel B: Large banks (total assets > \$1 billion) (N=427), based on coefficients from Table 4</i>				
Actual change (\$Bn)	168	121	-52	-503
<i>Illiquid Assets</i> adjustment (\$Bn)	-8***	-4**	3**	3**
<i>Core Deposits</i> adjustment (\$Bn)	-5	3	51***	53***
<i>Capital</i> adjustment (\$Bn)	6	-2	11*	13**
<i>Commit</i> adjustment (\$Bn)	-86*	-56	-60	168***
Adjusted change (\$Bn)	75	62	-47	-265

Table 6: economic magnitudes

6.6 Table 7: loan-demand robustness

Read:

- Table 7 adds controls for loan shares interacted with the TED spread and state-by-TED effects.
- The main coefficient patterns remain similar to the base model.
- Large-bank results continue to show that illiquid assets and commitments reduce credit supply under stress, while stable funding supports credit supply.

Economic message: The robustness tests reduce concerns that the baseline results are only driven by regional or sectoral differences in borrower loan demand.

Table 7

Fixed effects regressions of credit supply growth on TED spread, firm characteristics, and interactions, with loan demand controls.

This table reports fixed effects regressions of quarterly growth in credit standardized by beginning of period credit. The data are observed quarterly for a panel of US commercial banks over the period 2006Q1 through 2009Q2. Large banks are those banks with beginning of quarter assets greater than \$1 billion, and small banks are those banks with beginning of quarter assets less than \$1 billion. The first column replicates results from Table 3 for comparison. Column 2 adds the share of commercial and industrial loans, the share of loans in real estate, and their interactions with TED to sweep out potential demand effects. Column 3 instead introduces state fixed effects based on the bank's headquarters and interactions between these state effects and TED. Standard errors, clustered at the firm level, are reported in parentheses. ***, **, and * denote that the coefficients are statistically significantly different from zero at the 1%, 5%, and 10% level, respectively.

	Base model (Table 3, Column 4)	With loan shares TED	With state TED effects
	(1)	(2)	(3)
<i>Panel A: Large banks (total assets > \$1 billion)</i>			
Illiquid Assets/Assets _{t-1}	0.030*	0.027	0.038**
	(0.017)	(0.017)	(0.017)
Illiquid Assets/Assets _{t-1} *TED	-1.340*	-0.897	-1.787**
	(0.762)	(0.781)	(0.791)
Core Deposits/Assets _{t-1}	-0.044***	-0.049***	-0.045***
	(0.011)	(0.012)	(0.012)
Core Deposits/Assets _{t-1} *TED	1.356**	1.594***	0.940
	(0.555)	(0.567)	(0.594)
Capital/Assets _{t-1}	0.088*	0.091*	0.095*
	(0.054)	(0.055)	(0.055)
Capital/Assets _{t-1} *TED	5.984**	5.581**	5.093*
	(2.699)	(2.626)	(2.847)
Commit/(Commit + Assets) _{t-1}	-0.012	-0.009	-0.019
	(0.032)	(0.031)	(0.032)
Commit/(Commit + Assets) _{t-1} *TED	-2.753***	-3.676***	-2.613***
	(0.855)	(0.869)	(1.006)
Log Assets _{t-1}	-0.027***	-0.028***	-0.027***
	(0.007)	(0.007)	(0.007)
Log Assets _{t-1} *TED	0.131*	0.112*	0.234***
	(0.067)	(0.067)	(0.081)
Firm dummies	Yes	Yes	Yes
Quarterly time dummies	Yes	Yes	Yes
Loan share and loan shares TED	No	Yes	No
State dummies TED	No	No	Yes
N	6,462	6,462	6,462
Adjusted R-squared	0.342	0.345	0.347
<i>Panel B: Small banks (total assets ≤ \$1 billion)</i>			
Illiquid Assets/Assets _{t-1}	-0.051***	-0.058***	-0.049***
	(0.006)	(0.005)	(0.006)
Illiquid Assets/Assets _{t-1} *TED	-0.516***	-0.335*	-0.504***
	(0.183)	(0.201)	(0.187)
Core Deposits/Assets _{t-1}	-0.002	-0.008	0.001
	(0.006)	(0.006)	(0.006)
Core Deposits/Assets _{t-1} *TED	1.553***	1.394***	1.155***
	(0.238)	(0.241)	(0.242)
Capital/Assets _{t-1}	0.067***	0.056***	0.069***
	(0.012)	(0.010)	(0.012)
Capital/Assets _{t-1} *TED	1.058***	1.075***	0.915***
	(0.325)	(0.343)	(0.328)
Commit/(Commit + Assets) _{t-1}	-0.137***	-0.144***	-0.138***
	(0.012)	(0.012)	(0.012)
Commit/(Commit + Assets) _{t-1} *TED	-3.034***	-2.881***	-3.631***
	(0.557)	(0.553)	(0.581)
Log Assets _{t-1}	-0.028***	-0.038***	-0.027***
	(0.006)	(0.002)	(0.006)
Log Assets _{t-1} *TED	0.021	0.030	0.063**
	(0.024)	(0.030)	(0.026)
Firm dummies	Yes	Yes	Yes
Quarterly time dummies	Yes	Yes	Yes
Loan share and loan shares TED	No	Yes	No
State dummies TED	No	No	Yes
N	78,581	78,581	78,581
Adjusted R-squared	0.299	0.303	0.302

7 Short summary

- The paper studies how banks managed liquidity risk during the 2007–2009 financial crisis.
- It uses quarterly Call Report data from 2006Q1 to 2009Q2.
- The key crisis shock is the TED spread, with a crisis dummy as an alternative measure.
- Banks with more illiquid assets increase liquid assets and reduce credit supply when liquidity stress rises.
- Banks with more core deposits and equity capital maintain or expand lending relative to other banks.
- Off-balance sheet commitments become a liquidity problem when borrowers draw them down.
- The results are stronger and more economically important for large banks.
- Economic magnitude calculations imply that liquidity exposure explains a large share of credit supply contraction in 2008Q4.
- The paper shows that modern bank liquidity risk comes from both sides of the balance sheet and from off-balance sheet commitments.

8 Conclusion

8.1 Core conclusion

Cornett, McNutt, Strahan, and Tehranian show that liquidity risk management at banks contributed to the contraction of credit during the financial crisis. Banks with vulnerable balance sheets hoarded liquidity and reduced credit supply, while banks with stable funding sources and stronger capital were better able to lend.

8.2 Economic intuition

The crisis created a system-wide liquidity shock, but the shock affected banks differently. Banks with illiquid assets and large unused commitments needed to protect themselves by accumulating liquid assets. Banks funded by stable core deposits and capital faced less pressure. As a result, differences in pre-crisis balance sheets translated into differences in lending during the crisis.

8.3 Takeaway

The paper's main lesson is that bank liquidity risk management is a real credit-supply channel. Liquidity regulation should focus not only on cash holdings but also on asset liquidity, funding stability, capital, and off-balance sheet commitments.